SHIRWIN PRINCE I

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PROFESSIONAL SUMMARY

Driven **AI and Machine Learning Engineer** and **Software Systems** graduate with hands-on experience in building and deploying intelligent systems. Specialized in **Python**, **Deep Learning**, and **NLP**, with a proven track record of developing end-to-end solutions—from RAG-based chatbots to computer vision applications. Expert in data preprocessing, feature engineering, and model deployment using **Flask** and **Streamlit** to solve real-world healthcare and business problems.

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Machine Learning:** Regression, Classification, Clustering (K-Means, DBSCAN), Feature Engineering, Model Evaluation
- **Deep Learning & NLP:** CNNs, NLP, Sentiment Analysis, Logistic Regression, RAG-based Chatbots
- **Libraries & Frameworks:** Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, TensorFlow, Keras
- **Tools & Platforms:** Flask, Streamlit, Power BI (DAX, DAX Studio), Git, GitHub, Azure ML

PROJECTS

Sign Language Detection – Multi-Model Computer Vision Application
Tech Stack: Python, TensorFlow, MediaPipe, CNN, LSTM, OpenCV, NumPy

- **Developed** a multi-stage recognition system utilizing hybrid architectures to classify hand gestures into numbers, alphabets, and words with high precision.
- **Implemented** a real-time **MediaPipe** landmark detection pipeline for numerical recognition, leveraging geometric hand-point analysis for direct classification.
- **Architected** a **Convolutional Neural Network (CNN)** to perform spatial feature extraction and image classification for manual alphabet gestures.
- **Engineered** a **Long Short-Term Memory (LSTM)** network to process temporal sequences of MediaPipe landmarks, enabling accurate recognition of dynamic, word-level sign language.
- **Optimized** system performance through extensive data augmentation (rotation, scaling, and lighting shifts) to ensure robust detection across diverse environments

EyeVision AI – Deep Learning Eye Disease Classification Platform
Tech Stack: Python, Deep Learning, ResNet50, Flask, React.js, Tailwind CSS, SQLite

- **Architected** a full-stack medical imaging application utilizing a fine-tuned **ResNet50 CNN** to classify retinal fundus images into four categories (Cataract, Diabetic Retinopathy, Glaucoma, and Normal) with high accuracy.
- **Developed** a robust backend using **Flask** and **SQLite**, implementing role-based authentication and a relational database to manage appointments and doctor/patient portals.
- **Engineered** a responsive frontend with **React.js** and **Tailwind CSS**, featuring real-time interactive health chatbots and **Framer Motion** animations for enhanced UX.
- **Integrated** automated reporting workflows using **ReportLab** to generate and share AI-driven PDF care plans via email, bridging the gap between automated diagnosis and clinical management.

Sentifyy – AI-Powered Sentiment Analysis Platform
Tech Stack: Python, NLP, Logistic Regression, Scikit-learn, Flask, RAG, LLM

- **Engineered** an NLP-based sentiment classification system utilizing **Logistic Regression** and custom text preprocessing pipelines to achieve **90% accuracy** on user input datasets.
- **Developed and Integrated** a **Retrieval-Augmented Generation (RAG)** chatbot to enhance user engagement by providing context-aware, LLM-driven responses.
- **Deployed** the trained models via a **Flask** web interface, enabling low-latency, real-time sentiment predictions and interactive AI communication.

Bone Fracture Detection – Deep Learning Medical Imaging System
Tech Stack: Python, Deep Learning, CNN, TensorFlow, Keras, OpenCV

- **Architected and Trained** a **Convolutional Neural Network (CNN)** using **TensorFlow/Keras** to automate bone fracture detection in X-ray imagery, achieving **89% diagnostic accuracy**.
- **Optimized** model generalization by implementing advanced **image preprocessing** and **data augmentation** pipelines (rotation, scaling, and contrast adjustment) to handle diverse clinical datasets.
- **Engineered** a seamless integration between the trained model and a diagnostic interface, streamlining the healthcare workflow to support faster and more reliable **clinical decision-making**.

Retail Price Analyzer – Predictive Pricing System
Tech Stack: Python, Decision Tree Regression, Scikit-learn, Streamlit, Pandas

- **Implemented** a high-performance **Decision Tree Regression** model to forecast product prices, achieving **94% accuracy** through the analysis of historical retail datasets.
- **Optimized** model performance by executing rigorous **data preprocessing** and **feature engineering**, identifying the most influential variables for price volatility.
- **Developed** and deployed an interactive **Streamlit** application, enabling stakeholders to conduct real-time price analysis and execute data-driven pricing strategies.

Stock Keeping Unit (SKU) Clustering – Unsupervised ML Optimization
Tech Stack: Python, Unsupervised Learning, K-Means, DBSCAN, Matplotlib, Seaborn

- **Applied** unsupervised machine learning algorithms, including **K-Means** and **DBSCAN**, to segment SKUs based on complex inventory turnover and sales velocity patterns.
- **Achieved** an **85% clustering accuracy**, identifying high-value product groups to directly support **demand forecasting**, **inventory optimization**, and profit scalability.
- **Synthesized** clustering results into actionable visualizations, providing strategic insights that streamlined supply chain planning and reduced stock-out risks

EXPERIENCE

May 2023 – July 2023

- Supported logistics operations by managing IT assets related to stock management and delivery workflows.
- Utilized SQL, Excel, and Power BI to analyze operational data and improve reporting efficiency.
- Assisted in user training and technical support, contributing to improved system adoption and workflow efficiency.

EDUCATION

M.Sc. Software Systems | KG College of Arts and Science | 2021 – Present — CGPA: 8.62 / 10
Higher Secondary Certificate | CSI Matriculation Higher Secondary School | 2021 — Percentage: 80%

CERTIFICATIONS

- **Data Science for Python**– Appin Technology (2026)
- **Python for Data Science, AI & Development** – Coursera (2025)
- **Microsoft Azure AI Fundamentals (AI-900)** – Coursera (2025)
- **Generative AI Assistants** – Coursera (2025)
- **Mastering SQL Joins** – Coursera (2023)
- **Crash Course on Python** – Coursera (2022)
- **Programming Foundations with JavaScript, HTML and CSS –** Coursera (2022)

SOFT SKILLS:
Analytical Thinking | Problem Solving | Adaptability | Team Collaboration | Quick Learner

LANGUAGE:

English (Conversational) | Hindi (Basic) | Malayalam (Basic) | Tamil (Native)